

Occupational Safety and Health (Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013

**SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING**
**Product Identifier**

Perihalan Produk: **Phosphorus tribromide**  
 Product Description: **Phosphorus tribromide**  
 Cat No. : C16947  
 Synonyms Tribromophosphine.  
 CAS No 7789-60-8  
 Molecular Formula Br<sub>3</sub> P

**Relevant identified uses of the substance or mixture and uses advised against**

Recommended Use Laboratory chemicals.  
 Uses advised against No Information available

**Company**

Thermo Fisher Scientific Fisher Scientific (M) Sdn Bhd  
 Hap Seng Business Park, Lot 01-03, 01-04 Aras 1 Unity Square,  
 No 12, Persiaran Perusahaan, Seksyen 23, 40300 Shah Alam,  
 Selangor Darul Ehsan, Malaysia.  
 Main line: +60 3-5525 7888

**Supplier**

E-mail address Enquiry.my@thermofisher.com

**Emergency Telephone Number**

Tel: +03-5525 7888  
 CHEMTREC Malaysia **1-800-815-308** (Malay)  
 CHEMTREC Malaysia (Kuala Lumpur) **+(60)-327884561** (Malay)

**SECTION 2: HAZARDS IDENTIFICATION**
**Classification of the substance or mixture**

|                                                    |                     |
|----------------------------------------------------|---------------------|
| Skin Corrosion/Irritation                          | Category 1 B (H314) |
| Serious Eye Damage/Eye Irritation                  | Category 1 (H318)   |
| Specific target organ toxicity - (single exposure) | Category 3 (H335)   |

**Label Elements**


Signal Word

Danger

# SAFETY DATA SHEET

Phosphorus tribromide

Revision Date 31-Mar-2025

## Hazard Statements

H314 - Causes severe skin burns and eye damage  
H335 - May cause respiratory irritation

## Precautionary Statements

### Prevention

P264 - Wash face, hands and any exposed skin thoroughly after handling  
P271 - Use only outdoors or in a well-ventilated area  
P280 - Wear protective gloves/protective clothing/eye protection/face protection

### Response

P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P303 + P361 + P353 - IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water or shower  
P304 + P340 - IF INHALED: Remove person to fresh air and keep comfortable for breathing  
P305 + P351 + P338 - IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
P310 - Immediately call a POISON CENTER or doctor  
P362 + P364 - Take off contaminated clothing and wash it before reuse

### Storage

P403 + P233 - Store in a well-ventilated place. Keep container tightly closed

### Disposal

P501 - Dispose of contents/ container to an approved waste disposal plant

## Other Hazards

EUH014 - Reacts violently with water

This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

| Component             | CAS No    | Weight % |
|-----------------------|-----------|----------|
| Phosphorus tribromide | 7789-60-8 | >95      |

## SECTION 4: FIRST AID MEASURES

### Description of first aid measures

#### Eye Contact

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### Skin Contact

Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes.

#### Ingestion

Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water.

#### Inhalation

Remove from exposure, lie down. Remove to fresh air.

#### Self-Protection of the First Aider

Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination.

### Most important symptoms and effects, both acute and delayed

Causes burns by all exposure routes. . Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation.

# SAFETY DATA SHEET

Phosphorus tribromide

Revision Date 31-Mar-2025

## Indication of any immediate medical attention and special treatment needed

### Notes to Physician

Treat symptomatically.

## SECTION 5: FIREFIGHTING MEASURES

### Extinguishing media

#### Suitable Extinguishing Media

Do not use a solid water stream as it may scatter and spread fire.

#### Extinguishing media which must not be used for safety reasons

No information available.

### Special hazards arising from the substance or mixture

Reacts violently with water.

### Hazardous Combustion Products

Hydrogen halides, Oxides of phosphorus.

### Advice for fire-fighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### Personal Precautions, Protective Equipment and Emergency Procedures

Ensure adequate ventilation.

### Environmental precautions

See Section 12 for additional Ecological Information.

### Methods and Material for Containment and Cleaning Up

Soak up with inert absorbent material (e.g. sand, silica gel, acid binder, universal binder, sawdust). Keep in suitable, closed containers for disposal. Do not expose spill to water.

### Reference to Other Sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### Precautions for Safe Handling

Do not breathe mist/vapors/spray. Do not get in eyes, on skin, or on clothing. Handle product only in closed system or provide appropriate exhaust ventilation. Handle under inert gas, protect from moisture. Do not allow contact with water. Do not allow contact with water because of violent reaction.

### Conditions for Safe Storage, Including any Incompatibilities

Keep in a dry, cool and well-ventilated place. Keep container tightly closed. Protect from moisture. Keep from any possible contact with water. Corrosives area.

### Specific End Uses

ALFAAC16947

# SAFETY DATA SHEET

Phosphorus tribromide

Revision Date 31-Mar-2025

Use in laboratories.

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### Control Parameters

#### Exposure Controls

##### **Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

#### Personal protective equipment

##### **Eye Protection**

Goggles

##### **Hand Protection**

Protective gloves

##### **Skin and body protection**

Wear appropriate protective gloves and clothing to prevent skin exposure

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

##### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators

##### **Recommended Filter type:**

Particulates filter conforming to EN 143 Inorganic gases and vapours filter Type B Grey conforming to EN14387

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

When RPE is used a face piece Fit Test should be conducted

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice

#### Environmental exposure controls

No information available

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

#### Information on basic physical and chemical properties

##### **Appearance**

Light yellow

##### **Physical State**

Liquid

##### **Odor**

Strong

##### **Odor Threshold**

No data available

##### **pH**

No information available

##### **Melting Point/Range**

-40 °C / -40 °F

##### **Softening Point**

No data available

# SAFETY DATA SHEET

Phosphorus tribromide

Revision Date 31-Mar-2025

|                                         |                          |                                   |
|-----------------------------------------|--------------------------|-----------------------------------|
| Boiling Point/Range                     | 175 °C / 347 °F          |                                   |
| Flash Point                             | No information available | Method - No information available |
| Evaporation Rate                        | No data available        |                                   |
| Flammability (solid,gas)                | Not applicable           | Liquid                            |
| Explosion Limits                        | No data available        |                                   |
| Vapor Pressure                          | 10 mmHg @ 47.8 °C        |                                   |
| Vapor Density                           | 9.3                      | (Air = 1.0)                       |
| Specific Gravity / Density              | 2.880                    |                                   |
| Bulk Density                            | Not applicable           | Liquid                            |
| Water Solubility                        | reacts                   |                                   |
| Solubility in other solvents            | No information available |                                   |
| Partition Coefficient (n-octanol/water) |                          |                                   |
| Component                               | log Pow                  |                                   |
| Phosphorus tribromide                   | 2.28                     |                                   |
| Autoignition Temperature                | No data available        |                                   |
| Decomposition Temperature               | No data available        |                                   |
| Viscosity                               | No data available        |                                   |
| Explosive Properties                    | No information available |                                   |
| Oxidizing Properties                    | No information available |                                   |
| Molecular Formula                       | Br3 P                    |                                   |
| Molecular Weight                        | 270.69                   |                                   |

## SECTION 10: STABILITY AND REACTIVITY

### Reactivity

Yes.

### Chemical Stability

Moisture sensitive.

### Possibility of Hazardous Reactions

|                          |                           |
|--------------------------|---------------------------|
| Hazardous Polymerization | No information available. |
| Hazardous Reactions      | No information available. |

### Conditions to Avoid

Incompatible products. Exposure to moist air or water.

### Incompatible Materials

Acids. Strong oxidizing agents. Strong bases. Alcohols. Ammonia. Metals.

### Hazardous Decomposition Products

# SAFETY DATA SHEET

Phosphorus tribromide

Revision Date 31-Mar-2025

Hydrogen halides. Oxides of phosphorus.

## SECTION 11: TOXICOLOGICAL INFORMATION

### Information on Toxicological Effects

|                                                  |                                                                                                                                                                                                                                                             |
|--------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Product Information</b>                       | No acute toxicity information is available for this product                                                                                                                                                                                                 |
| <b>(a) acute toxicity;</b>                       |                                                                                                                                                                                                                                                             |
| Oral                                             | No data available                                                                                                                                                                                                                                           |
| Dermal                                           | No data available                                                                                                                                                                                                                                           |
| Inhalation                                       | No data available                                                                                                                                                                                                                                           |
| <b>(b) skin corrosion/irritation;</b>            | Category 1 B                                                                                                                                                                                                                                                |
| <b>(c) serious eye damage/irritation;</b>        | Category 1                                                                                                                                                                                                                                                  |
| <b>(d) respiratory or skin sensitization;</b>    |                                                                                                                                                                                                                                                             |
| Respiratory                                      | No data available                                                                                                                                                                                                                                           |
| Skin                                             | No data available                                                                                                                                                                                                                                           |
| <b>(e) germ cell mutagenicity;</b>               | No data available                                                                                                                                                                                                                                           |
| <b>(f) carcinogenicity;</b>                      | No data available                                                                                                                                                                                                                                           |
|                                                  | There are no known carcinogenic chemicals in this product                                                                                                                                                                                                   |
| <b>(g) reproductive toxicity;</b>                | No data available                                                                                                                                                                                                                                           |
| <b>(h) STOT-single exposure;</b>                 | Category 3                                                                                                                                                                                                                                                  |
| Results / Target organs                          | Respiratory system.                                                                                                                                                                                                                                         |
| <b>(i) STOT-repeated exposure;</b>               | No data available                                                                                                                                                                                                                                           |
| Target Organs                                    | No information available.                                                                                                                                                                                                                                   |
| <b>(j) aspiration hazard;</b>                    | No data available                                                                                                                                                                                                                                           |
| <b>Other Adverse Effects</b>                     | See actual entry in RTECS for complete information                                                                                                                                                                                                          |
| <b>Symptoms / effects,both acute and delayed</b> | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated. Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation. |
| <b>Endocrine Disrupting Properties</b>           | Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.                                                                                                                         |

## SECTION 12: ECOLOGICAL INFORMATION

# SAFETY DATA SHEET

Phosphorus tribromide

Revision Date 31-Mar-2025

**Ecotoxicity effects** Do not empty into drains.

**Persistence and degradability**  
**Persistence** Soluble in water, Persistence is unlikely, based on information available.  
**Degradability** Not relevant for inorganic substances.

**Bioaccumulative potential** Bioaccumulation is unlikely

| Component             | log Pow | Bioconcentration factor (BCF) |
|-----------------------|---------|-------------------------------|
| Phosphorus tribromide | 2.28    | No data available             |

**Mobility in soil** The product is water soluble, and may spread in water systems. Will likely be mobile in the environment due to its water solubility. Highly mobile in soils.

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

**Other adverse effects** No information available

## SECTION 13: DISPOSAL CONSIDERATIONS

**Waste treatment methods**  
**Waste from Residues/Unused Products** Waste is classified as hazardous Dispose of in accordance with the European Directives on waste and hazardous waste Dispose of in accordance with local regulations

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used Do not empty into drains Do not flush to sewer Large amounts will affect pH and harm aquatic organisms

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO**  
**UN-No** UN1808  
**Hazard Class** 8  
**Packing Group** II  
**Proper Shipping Name** PHOSPHORUS TRIBROMIDE

**Road and Rail Transport**  
**UN-No** UN1808  
**Hazard Class** 8  
**Packing Group** II  
**Proper Shipping Name** PHOSPHORUS TRIBROMIDE

**IATA**  
**UN-No** UN1808  
**Hazard Class** 8  
**Packing Group** II  
**Proper Shipping Name** PHOSPHORUS TRIBROMIDE

# SAFETY DATA SHEET

Phosphorus tribromide

Revision Date 31-Mar-2025

**Special Precautions for User** No special precautions required

## SECTION 15: REGULATORY INFORMATION

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

**International Inventories** X = listed

| Component             | EINECS    | TSCA | DSL | PICCS | ENCS | ISHL | IECSC | AICS | KECL     |
|-----------------------|-----------|------|-----|-------|------|------|-------|------|----------|
| Phosphorus tribromide | 232-178-2 | X    | X   | X     | X    | X    | X     | X    | KE-28722 |

### National Regulations

**Persistent Organic Pollutant** This product does not contain any known or suspected substance  
**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 16: OTHER INFORMATION

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**POW** - Partition coefficient Octanol:Water

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### **Key literature references and sources for data**

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**Prepared By**

**Revision Date**

**Revision Summary**

Health, Safety and Environmental Department

31-Mar-2025

Not applicable.

**In accordance with local and national regulations: Occupational Safety and Health**



# SAFETY DATA SHEET

Phosphorus tribromide

Revision Date 31-Mar-2025

---

**(Classification, Labelling and Safety Data Sheet of Hazardous Chemicals) Regulations 2013**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**